Muhammad Hassan Rajput

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GitHub | LinkedIn | Portfolio

SKILLS

Data Processing & Visualization

- **NumPy, Pandas**
- **Matplotlib, Seaborn**

Machine Learning & Computer Vision

- **TensorFlow, Keras**
- **Scikit-Learn**
- **OpenCV**

Web Development

- **HTML, CSS / Tailwind**
- **MERN Stack, MySQL**
- **Next.js**
- **FastAPI**
- **Streamlit**

Tools

- **Git / GitHub**
- **Huggingface**

PROJECTS

KOA Detection System – TensorFlow, Keras, OpenCV, React , fastapi

- Achieved **83.78% validation accuracy** and **92.1% training accuracy** on 3-class KL grading (KL0–1, KL2–3, KL4) by fine-tuning a pretrained **DenseNet121** in TensorFlow/Keras.
- Achieved **0.995 mAP@50** knee localization by training a YOLOv8 ROI detector with OpenCV to crop knee regions before KL grading.
- Delivered a full-stack web app serving KL-grade predictions with ScoreCAM heatmaps by building a **React** + Tailwind frontend and a **FastAPI** backend with Pydantic validation and MongoDB storage.

Nail Disease Detection System – TensorFlow, Keras, React, FastAPI

- Achieved **98.64% validation accuracy** and **99.9% training accuracy** in binary nail-disease detection by training a CNN in TensorFlow/Keras.
- Achieved **91.07% validation accuracy** and **90.2% training accuracy** in multi-class nail-disease classification by training a second CNN, forming a two-stage pipeline.
- Enabled browser-based nail-disease diagnosis by building a **React** frontend that consumes a **FastAPI** backend serving both classifiers through REST endpoints.

Typing Clash – MERN stack, Tailwind CSS, Socket.io

- Achieved multiplayer functionality (4 users), as measured by socket sessions, by implementing real-time communication.
- Increased engagement by 40% during a hackathon via competitive, stats-based gameplay.

EDUCATION

- Mehran University of Engineering and Technology — *CGPA: 3.2|(20-may -2026)*
- XII Excellence College for boys — *87.8%| 2022*

EXPERIENCE

- Worked @Zetsol Tech — zetsol.co/ — (17 Aug – 25 Sep 2026)
- Interned @YoungDev — linkedin.com/company/youngdev-intern/ — (1 month)
- Web Director @IEEE CS Society — ieee.org/ — (4 months)